

# Assignment–4

## Design of Synchronous Mod-5 Counter using JK Flip-Flops

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### Aim

To design and implement a Synchronous Mod-5 Counter using JK Flip-Flops.

### Theoretical Background

A counter is a sequential circuit that changes state with each clock pulse.

A Mod- $N$  counter counts from 0 to  $(N - 1)$  and then repeats.

For a Mod-5 counter, the required number of flip-flops satisfies:

$$2^n \geq 5$$

$$2^2 = 4 (< 5), \quad 2^3 = 8 (\geq 5)$$

Therefore, a minimum of three flip-flops is required.

In a synchronous counter, all flip-flops are driven by a common clock, eliminating ripple delay.

### JK Flip-Flop Characteristic Table

| J | K | $Q_{next}$     |
|---|---|----------------|
| 0 | 0 | Q              |
| 0 | 1 | 0              |
| 1 | 0 | 1              |
| 1 | 1 | $\overline{Q}$ |

### JK Excitation Table

| Q | $Q_{next}$ | J | K |
|---|------------|---|---|
| 0 | 0          | 0 | X |
| 0 | 1          | 1 | X |
| 1 | 0          | X | 1 |
| 1 | 1          | X | 0 |

## State Sequence (Mod-5)

The required counting sequence is:

$$000 \rightarrow 001 \rightarrow 010 \rightarrow 011 \rightarrow 100 \rightarrow 000$$

| Decimal | $Q_2 Q_1 Q_0$ |
|---------|---------------|
| 0       | 000           |
| 1       | 001           |
| 2       | 010           |
| 3       | 011           |
| 4       | 100           |

## State Transition Table

| Present State ( $Q_2 Q_1 Q_0$ ) | Next State ( $Q_2^+ Q_1^+ Q_0^+$ ) |
|---------------------------------|------------------------------------|
| 000                             | 001                                |
| 001                             | 010                                |
| 010                             | 011                                |
| 011                             | 100                                |
| 100                             | 000                                |

## Complete Excitation Table for Mod-5 Counter

| $Q_2$ | $Q_1$ | $Q_0$ | $Q_2^+$ | $Q_1^+$ | $Q_0^+$ | $J_0$ | $K_0$ | $J_1$ | $K_1$ | $J_2$ | $K_2$ |
|-------|-------|-------|---------|---------|---------|-------|-------|-------|-------|-------|-------|
| 0     | 0     | 0     | 0       | 0       | 1       | 1     | X     | 0     | X     | 0     | X     |
| 0     | 0     | 1     | 0       | 1       | 0       | X     | 1     | 1     | X     | 0     | X     |
| 0     | 1     | 0     | 0       | 1       | 1       | 1     | X     | X     | 0     | 0     | X     |
| 0     | 1     | 1     | 1       | 0       | 0       | X     | 1     | X     | 1     | 1     | X     |
| 1     | 0     | 0     | 0       | 0       | 0       | 0     | 0     | 0     | X     | X     | 1     |

## Derived Input Equations Using K-Maps

Variables: Rows  $\rightarrow Q_2$

Columns (Gray order)  $\rightarrow Q_1 Q_0 = 00, 01, 11, 10$

**$J_0$  K-Map**

|   | 00 | 01 | 11 | 10 |
|---|----|----|----|----|
| 0 | 1  | X  | X  | 1  |
| 1 | 0  | X  | 0  | X  |

$$J_0 = \overline{Q_2}$$

 **$K_0$  K-Map**

|   | 00 | 01 | 11 | 10 |
|---|----|----|----|----|
| 0 | X  | 1  | 1  | X  |
| 1 | X  | 1  | 1  | X  |

$$K_0 = Q_0$$

 **$J_1$  K-Map**

|   | 00 | 01 | 11 | 10 |
|---|----|----|----|----|
| 0 | 0  | 1  | X  | X  |
| 1 | 0  | 0  | X  | X  |

$$J_1 = \overline{Q_2}Q_0$$

 **$K_1$  K-Map**

|   | 00 | 01 | 11 | 10 |
|---|----|----|----|----|
| 0 | X  | X  | 1  | 0  |
| 1 | X  | X  | 1  | 1  |

$$K_1 = Q_0 + Q_2$$

 **$J_2$  K-Map**

|   | 00 | 01 | 11 | 10 |
|---|----|----|----|----|
| 0 | 0  | 0  | 1  | 0  |
| 1 | X  | X  | X  | X  |

$$J_2 = Q_1Q_0$$

 **$K_2$  K-Map**

|   | 00 | 01 | 11 | 10 |
|---|----|----|----|----|
| 0 | X  | X  | X  | X  |
| 1 | 1  | 1  | 1  | 1  |

$$K_2 = 1$$

## Unused States

The unused states are: 101, 110, 111

These are treated as don't care conditions in K-map simplification. The reset logic ensures that if the counter enters 101, it returns to 000.

## Working of the Counter

Initially assume the state is 000.

With each clock pulse, the counter transitions as:

$$000 \rightarrow 001 \rightarrow 010 \rightarrow 011 \rightarrow 100 \rightarrow 000$$

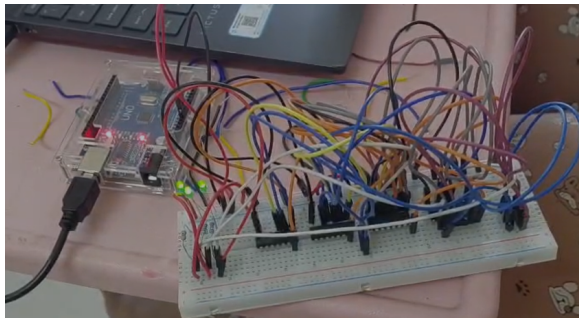
Thus, the counter cycles through five states and repeats, achieving Mod-5 operation.

## Circuit Connections

- The ICs used in the circuit are: U74HC08L (AND gates), SN74ALS02AN (NOR gates), and SN74LS76N / HD74LS76AP (JK flip-flops).
- All ICs are powered using +5V and GND from the Arduino board. Pin 16 of U74HC08L and SN74ALS02AN is connected to +5V and Pin 8 is connected to GND. Pin 5 (VCC) and Pin 13 (GND) of the SN74LS76N / HD74LS76AP are connected to +5V and GND respectively
- The clock input from the Arduino digital output pin is connected to the clock inputs of all JK flip-flops to ensure synchronous operation.
- The outputs of the JK flip-flops are connected to the inputs of the U74HC08L (AND gates) and SN74ALS02AN (NOR gates) to generate the required combinational logic.
- The combinational logic generates the required J and K inputs for each flip-flop according to the Mod-5 counting sequence.
- The three flip-flop outputs represent the counter bits:

$$Q_2, Q_1, Q_0$$

- A reset condition is implemented using the NOR gate to detect the state 101 (decimal 5). When this state occurs, the flip-flops are cleared to 000.
- All unused inputs are properly tied to either HIGH or LOW to prevent floating conditions.



## Observations

### Improper Reset Handling

At the beginning, the counter sometimes started from a random value instead of 000. This happened because the flip-flops were not reset properly before giving the clock signal.

**Precaution:** Always give a clear/reset pulse before starting the clock so that the counter begins from 000.

### Unused States Causing Glitches

The unused states (101, 110, 111) gave unexpected outputs during testing. This was because the don't-care conditions were not handled properly in the logic design.

**Precaution:** Check the K-map grouping carefully and make sure the logic expressions are simplified correctly.

### Clock Signal Noise

Sometimes the counter did not count properly because the clock signal from the Arduino was not perfectly stable. Small noise in the clock caused extra unwanted changes.

**Precaution:** Use a stable clock signal and make sure the grounding is proper to avoid false triggering.

## Loose Breadboard Connections

Some wrong outputs were seen because a few jumper wires were loose on the breadboard.

**Precaution:** Make sure all wires are properly connected and double-check the circuit before turning it on.

## Propagation Delay Between Logic Gates

There was a small delay in the LED output because of the propagation delay in the logic gates and JK flip-flops. However, the counting sequence was still correct.

**Precaution:** Use ICs from the same family and operate the circuit within the recommended frequency range.